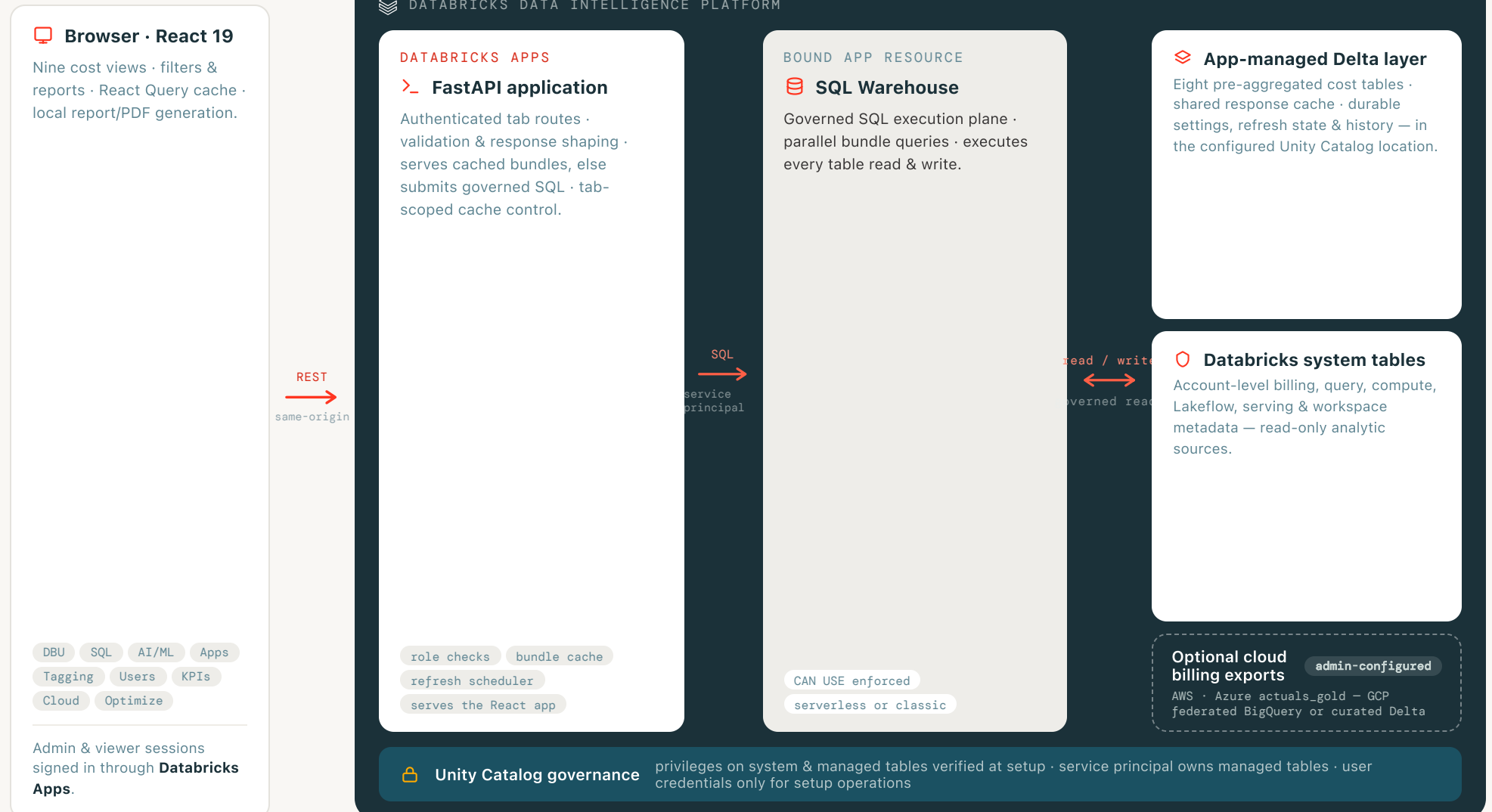




The browser calls a FastAPI service, which executes governed SQL through a bound Databricks SQL Warehouse and serves results from **system tables** or **app-managed Delta aggregates**. Static design information only — no customer, account, workspace, or user identifiers.

CLIENT



AUTHENTICATION + GOVERNANCE

Apps session → FastAPI role checks → service principal / setup user → Warehouse CAN USE + Unity Catalog grants.

SCHEDULED REFRESH

Scheduler → service principal → incremental Delta MERGEs → refresh state & history → cache invalidation.

ON-DEMAND REFRESH

Tab refresh → scoped cache clear → active refetch. Administrator rebuild → full aggregate recreation.

Request & data flow

- 1 A signed-in user opens the React interface in Databricks Apps and selects dates, workspaces, or report options.
- 2 The browser sends same-origin requests to tab-specific FastAPI routes.
- 3 FastAPI serves a valid cached bundle when possible; otherwise it submits governed SQL through the bound SQL Warehouse.
- 4 Queries use app-managed Delta aggregates for supported summaries and live system-table queries for detailed, specialized, or fallback views.
- 5 Configured AWS / Azure / GCP billing exports are queried only for the optional actual-cloud-cost views.
- 6 FastAPI returns shaped JSON; React renders the dashboard or generates the architecture PDF locally.

Authentication & governance

- **Databricks Apps** authenticates the browser session and forwards user identity to the app.
- **FastAPI roles** protect settings mutations and rebuilds; user-scoped credentials are reserved for setup operations that need the caller's privileges.
- The **app service principal** performs dashboard SQL and owns managed tables, scheduled refreshes, and cache writes.
- **Unity Catalog** privileges govern table access; the bound **SQL Warehouse** separately enforces CAN USE.

Refresh paths

Scheduled aggregate refresh nightly default

- 1 acquire the shared refresh lock
- 2 MERGE recent partitions into the 8 aggregates
- 3 save watermarks & history · rebuild source views
- 4 invalidate shared response caches

Weekly or monthly cadence selectable.

Administrator full rebuild Settings → Config → Rebuild

- 1 verify app administrator · queue locked rebuild
- 2 recreate all aggregates for the history window
- 3 write progress to refresh state & history
- 4 clear query + response caches on completion

On-demand tab refresh no rebuild

- 1 POST the tab name to /api/cache/clear
- 2 clear only that tab's process + shared cache
- 3 React Query refetches the active routes
- 4 aggregate tables stay unchanged

COMPONENTS SUMMARY

React browser interface

Interactive cost views, filters, settings, report downloads.

Databricks system tables

Billing, query, compute, Lakeflow, serving, workspace metadata.

FastAPI application

Validates requests, coordinates queries, shapes responses.

App-managed Delta layer

Aggregates, settings, refresh state, shared response cache.

Databricks SQL Warehouse

Governed SQL execution plane for every read and write.

Optional cloud exports

AWS / Azure / GCP actuals supplement Databricks charges.

Tab-by-tab data lineage

Tab	FASTAPI ROUTES (GET /API/...)	MANAGED DELTA / CACHE	SOURCE SYSTEM TABLES	FALLBACK / OPTIONAL ENRICHMENT
DBU Overview	billing:dashboard-fast · sku · interactive · pipelines · kpi	daily_usage_summary · daily_product_breakdown · app_response_cache	billing.usage · billing.list_prices · access.workspaces_latest · compute.clusters	Live billing scan if aggregates are empty; detail stays live. Opt: pipeline names + account prices.
SQL	billing:sql · dbsql:dashboard · top · top-by-source · by-user · platform-kpi	sql_tool_attribution · dbsql_cost_per_query · app_response_cache	billing.usage · billing.list_prices · query.history · compute.warehouses · access.workspaces_latest	Live billing + query-history fallback; per-query cost unavailable until its Delta table exists — no invented attribution.
AI/ML	aiml:dashboard · billing:kpi	app_response_cache (analytics live)	billing.usage · billing.list_prices	Billing-only views survive. Opt: compute, serving & workspace metadata enrich names.
Apps	apps:dashboard · apps:kpi	daily_apps_summary · app_response_cache	billing.usage · billing.list_prices · access.workspaces_latest	Live billing fallback; Apps API enriches names/resources, stale metadata tolerated.
Tagging	tagging:dashboard · top-by-tag · billing:kpi	daily_tag_summary · daily_usage_summary · app_response_cache	billing.usage · billing.list_prices	Totals fall back to live billing; untagged rows stay live. Opt: compute / Lakeflow names.
Users	users-groups:bundle	app_response_cache (analytics live)	billing.usage · billing.list_prices	Opt: workspace SCIM APIs enrich billing identities with group membership.
KPIs & Trends	billing:kpis · billing:platform-kpi	daily_usage_summary · daily_query_stats · daily_workspace_breakdown · dbsql_cost_per_query · app_response_cache	billing.usage · billing.list_prices · query.history · lakeflow.job_run_timeline	Billing KPIs survive missing Lakeflow access; query-history trends remain live.
Cloud Costs	billing:infra · aws-actual:dashboard · azure-actual:dashboard · gcp-actual:dashboard · billing:kpi	app_response_cache (DBU/cluster live)	billing.usage · compute.clusters · access.workspaces_latest	DBU/cluster views work without exports; currency costs require a configured cloud export.
Optimize	sql:warehouse-health · warehouse-health/idle	none — 30-min in-process cache	compute.warehouses · compute.warehouse_events · query.history · billing.usage · billing.list_prices	Idle uptime can use billing intervals; rightsizing unavailable without regional tables.

Core analytic system tables

system.billing.usage
system.billing.list_prices
system.query.history
system.compute.clusters
system.compute.warehouses
system.compute.warehouse_events

system.lakeflow.jobs
system.lakeflow.pipelines
system.lakeflow.job_run_timeline
system.serving.served_entities
system.access.workspaces_latest

Optional: billing.account_prices · probe-only: access.audit (readiness check, never an analytic input).

App-managed Delta — configured UC location

8 AGGREGATES

daily_usage_summary
daily_product_breakdown
daily_workspace_breakdown
daily_query_stats

daily_tag_summary
daily_apps_summary
sql_tool_attribution
dbsql_cost_per_query

STATE & CACHE (13)

app_settings
app_schedule_settings
app_refresh_log
app_cloud_connections
app_workspace_filter
app_user_permissions
app_mv_refresh_state

app_alert_thresholds
app_webhook_settings
app_pricing_settings
app_response_cache
app_mv_sources
app_unified_views

Optional cloud billing sources

AWS <catalog>.
<schema>.actuals_gold

Azure <catalog>.
<schema>.actuals_gold

GCP <catalog>.
<schema>.<table>

Administrator-configured. GCP may be a federated BigQuery billing export or a curated Delta table. Queried only for actual-cloud-cost views.

Full route & fallback detail: cost-obs-architecture.md